1 The strategy

We aim to evaluate a string amplitude (free energy) of the form

$$\mathcal{F} = \int_{\mathcal{M}} d\mu, \tag{1.1}$$

where $\mathcal{M}$ is a suitable moduli space (say of the genus g Riemann surface) and $d\mu$ is a positive measure. The idea is to compare it with

$$\mathcal{F}_0 = \int_{\mathcal{M}} d\mu_0 \tag{1.2}$$

of a solvable model, and express

$$\mathcal{F} = \mathcal{F}_0 \left\langle \frac{d\mu}{d\mu_0} \right\rangle_0, \tag{1.3}$$

where

$$\langle f \rangle_0 = \frac{\int_{\mathcal{M}} d\mu_0 \, f}{\int_{\mathcal{M}} d\mu_0} \tag{1.4}$$

can be numerically evaluated by Monte Carlo sampling of f over $\mathcal{M}$ with probability measure $d\mu_0$.

2 The moduli space

The moduli space $\mathcal{M}_g$ of genus g Riemann surface can be viewed as the base of the obvious fibration

$$\pi : \mathcal{M}_{g,1} \to \mathcal{M}_g, \tag{2.1}$$

and $\mathcal{M}_{g,1}$ can be parameterized via the moduli space $\mathcal{M}_{g,1}^{\text{comb}}$ of metric ribbon graphs, via Kontsevich's isomorphism

$$\mathcal{M}_{g,1} \times \mathbb{R}_+ \xrightarrow{\sim} \mathcal{M}_{g,1}^{\text{comb}}. \tag{2.2}$$

The explicit ribbon graph is constructed by finding the Strebel differential $\omega = \varphi(z) dz^2$, whose zeros of degree d corresponding to vertices of the graph with valence $d+2$. The defining property of ω is such that all of its horizontal trajectories, along which $\varphi(z(s))(\frac{dz}{ds})^2 \in \mathbb{R}_+$, are closed or end on a zero. The length of a closed horizontal trajectory γ is equal to

$$\ell(\gamma) = \oint_\gamma \sqrt{\varphi(z)} \, dz, \tag{2.3}$$

and is constant along a continuous family of closed horizontal trajectories. In the case of one puncture, all closed horizontal trajectories are homologous and enclose the puncture where $\varphi(z)$ has a double pole.

At a generic point in the moduli space, the ribbon graph in question has only cubic vertices, and Euler characteristic $V - E + F = 2 - 2g$ with $3V = 2E$ and $F = 1$. So the number of vertices is $V = 4g - 2$, and the number of edges is $E = 6g - 3$.

Consider the example of a 1-punctured torus, parameterized by the holomorphic coordinate $z \sim z + 2\pi \sim z + 2\pi\tau$ with the puncture at $z = 0$. The Strebel differential is of the form

$$\omega = \ell^2 \left(\mathcal{P}_\tau(z) - a\right) dz^2, \tag{2.4}$$

where $\mathcal{P}_\tau(z)$ is the Weierstrass elliptic function which has a double zero at $z = 0$,

$$\mathcal{P}_\tau(z) = \frac{1}{z^2} + \sum_{(m,n)\neq(0,0)} \left[\frac{1}{(z - 2\pi(m + n\tau))^2} - \frac{1}{(2\pi(m + n\tau))^2}\right]. \tag{2.5}$$

The constant a is determined by τ by the condition that the horizontal trajectories are closed.

Rather than mapping metric graph data to plumbing parameters and computing CFT correlators from conformal blocks, the strategy is to directly computing CFT observable in terms of the metric graph data. For instance, the torus partition function of a free boson X can be represented, modulo Weyl anomaly, as

$$\begin{aligned} Z(\theta_1, \theta_2) = \int [DX(\sigma)]_{0<\sigma<2\pi} \Psi_1[X(\sigma)] \prod_{0<\sigma<\theta_1} \delta(X(\sigma) - X(\pi + \theta_1 - \sigma)) \\ \times \prod_{\theta_1<\sigma<\theta_2} \delta(X(\sigma) - X(\pi + \theta_1 + \theta_2 - \sigma)) \prod_{\theta_2<\sigma<\pi} \delta(X(\sigma) - X(2\pi + \theta_2 - \sigma)). \end{aligned} \tag{2.6}$$

Here $0 < \theta_1 < \theta_2 < 2\pi$ are related to the lengths ℓ_1, ℓ_2, ℓ_3 of the metric graph by $\theta_1 = \pi \frac{\ell_1}{\ell_1+\ell_2+\ell_3}$ and $\theta_2 = \pi \frac{\ell_1+\ell_2}{\ell_1+\ell_2+\ell_3}$. $\Psi_1[X(\sigma)]$ is the wave functional of the identity operator. As in (2.8.23) of Polchinski,

$$\Psi_1[X(\sigma)] \propto \exp\left(-\frac{1}{\alpha'} \sum_{m\geq 1} m X_m X_{-m}\right), \tag{2.7}$$

where X_m is the m-th Fourier mode of $X(\sigma)$.

To obtain the genus 2 partition function, the ribbon graph at a generic point in the moduli space has 6 cubic vertices and 9 edges. The partition function can be obtained by integrating $\Psi_1[X(\sigma)]$ with the identification of X on pairs of 18 segments of the unit circle.

3 Application to $c = 1$ string theory

The idea is to take $d\mu_R$ to be the measure associated with the free energy of $c = 1$ string theory at inverse temperature $\beta = 2\pi R$, and compare the measure at different values of R. Explicitly, as in (4.97) of the string notes, we have on $\mathcal{M}_g$

$$d\mu_R = \prod_{k=1}^{3g-3} i d\tau^k \wedge d\bar{\tau}^k \left\langle \prod_{k=1}^{3g-3} \mathcal{B}_{\bar{\tau}^k} \mathcal{B}_{\tau^k} \right\rangle_\Sigma , \tag{3.1}$$

which is a positive measure. The correlator is evaluated for the matter CFT that consists of $c = 25$ Liouville theory and a compact boson of radius R. It follows that

$$\frac{d\mu_R}{d\mu_{R_0}} = \frac{Z_R(\Sigma)}{Z_{R_0}(\Sigma)}, \tag{3.2}$$

where $Z_R(\Sigma)$ is the partition function of the compact boson of radius R on the Riemann surface Σ. Note that the Weyl anomaly cancels between the numerator and the denominator and so the RHS is well-defined.

Typically, the RHS of (3.2) is not hard to compute. The main challenge is to find an algorithm for sampling $\mathcal{M}_g$, say with Metropolis algorithm, subject to the probability measure $d\mu_{R_0}$.

In the case of the genus 2 free energy, there is an expression in terms of the period matrix Ω_{IJ},

$$\mathcal{F}_2(R) = \int_{\mathcal{M}_2} |\Phi(\Omega) d\Omega_{11} d\Omega_{22} d\Omega_{12}|^2 \frac{V_{26}}{(4\pi^2\alpha')^{26} (\det \mathrm{Im}\Omega)^{13} Z_{26}(\Sigma)} Z_{1,R}(\Sigma) Z_L(\Sigma), \tag{3.3}$$

where $\Phi(\Omega) = 2\pi i/\chi_{10}(\Omega)$ as in (4.109) of the string notes, $Z_{26}(\Sigma)$ is the genus two partition function of 26 noncompact free bosons with target space volume V_{26}, $Z_{1,R}(\Sigma)$ is the partition function of a compact boson at radius R, and $Z_L(\Sigma)$ is the $c = 25$ Liouville CFT partition function. Each partition function is subject to the Weyl anomaly, which cancels in the combination $Z_{1,R}(\Sigma) Z_L(\Sigma)/Z_{26}(\Sigma)$.

4 Ψ_1 of Liouville CFT

The Euclidean action of Liouville theory is

$$S_L = \frac{1}{4\pi} \int d^2\sigma \sqrt{g} (g^{ab} \partial_a \partial_b \phi + QR\phi + 4\pi\mu e^{2b\phi}), \tag{4.1}$$

where $Q = b + b^{-1}$. On the cylinder, the Hamiltonian reads

$$H = \int d\sigma \left[\pi \Pi_\phi^2 + \frac{1}{4\pi} (\partial_\sigma \phi)^2 + : \mu e^{2b\phi} : \right], \tag{4.2}$$

where : : is the normal ordering defined via the mode expansion

$$\begin{aligned} \phi &= \sum_{n \in \mathbb{Z}} \frac{1}{\sqrt{2|n|}} (a_n + a_{-n}^\dagger) e^{in\sigma}, \\ \Pi_\phi &= \frac{-i}{2\pi} \sum_{n \in \mathbb{Z}} \sqrt{\frac{|n|}{2}} (a_n - a_{-n}^\dagger) e^{in\sigma}. \end{aligned} \tag{4.3}$$

The exactness of this Hamiltonian can be justified through its commutation relation with the Virasoro generators defined by the holomorphic stress-energy tensor of the form

$$T = - : \partial\phi\partial\phi : + Q\partial^2\phi. \tag{4.4}$$

We could consider alternatively the regularized Hamiltonian defined by discretizing space,

$$H = \sum_{i=1}^{N} \left[\frac{N}{2} \Pi(i)^2 + \frac{N}{2(2\pi)^2} (\phi(i+1) - \phi(i))^2 + \mu \frac{2\pi}{N} e^{2b\phi(i)} \right], \tag{4.5}$$

where $\phi(N+1) \equiv \phi(1)$, $[\phi(i), \Pi(j)] = i\delta_{ij}$, and seek the identity wave function $\Psi_1(\phi(1), \cdots, \phi(N))$.

Note however that Ψ_1 is not the ground state, but rather the analytic continuation of a scattering state of real Liouville momentum P to $P = i\frac{Q}{2}$. The energy of Ψ_1 on the cylinder is $H = -\frac{1}{12} - \frac{Q^2}{2}$. Moreover, Ψ_1 is not normalizable and blows up exponentially in the large field region $\phi \to -\infty$ as $\Psi_1 \sim e^{-\frac{Q}{2}\phi}$. This leads to a divergence in the zero mode integration that is likely cancelled by the Weyl anomaly due to the Weyl frame of the metric associated with the ribbon graph/Strebel differential.

5 The partition function of free boson CFT from Ψ_1

Discretizing the unit circle into L sites, we consider the approximation of (2.7),

$$\begin{aligned} \Psi_1[X] &\propto \exp\left[-\frac{1}{\pi\alpha' L} \sum_{m=1}^{L} \sin\frac{\pi m}{L} \sum_{k,\ell=1}^{L} X(k) X(\ell) e^{2\pi i \frac{m}{L}(k-\ell)} \right] \\ &= \exp\left[-\frac{1}{\pi\alpha' L} \sum_{k,n=1}^{L} X(n) X(n+k) \frac{\sin\frac{\pi}{L}}{\cos\frac{2\pi k}{L} - \cos\frac{\pi}{L}} \right]. \end{aligned} \tag{5.1}$$

We will assume that L is even. Next we split $\frac{L}{2} = \ell_1 + \ell_2 + \ell_3$, where each ℓ_i is a positive integer, and make the identification

$$\begin{aligned} &X(\tfrac{L}{2} + \ell_1 + 1 - n) = X(n), \quad 1 \le n \le \ell_1, \\ &X(\tfrac{L}{2} + 2\ell_1 + \ell_2 + 1 - n) = X(n), \quad \ell_1 + 1 \le n \le \ell_1 + \ell_2, \\ &X(L + \ell_1 + \ell_2 + 1 - n) = X(n), \quad \ell_1 + \ell_2 + 1 \le n \le \tfrac{L}{2}, \end{aligned} \tag{5.2}$$

as a discretized version of (2.6). Finally, we perform the integration with respect to the remaining independent variables $X(n)$, $1 \le n \le \frac{L}{2}$.

After imposing the constraint (5.2), we can write $\Psi_1[X]$ as proportional to

$$\exp\left[-\sum_{k,\ell=1}^{\frac{L}{2}} A_{k\ell} X(k) X(\ell)\right], \tag{5.3}$$

where the eigenvalues of the matrix $A_{k\ell}$ are positive except for a single zero. The integration over $X(k)$, $1 \le k \le \frac{L}{2}$ then produces, up to a constant normalization factor, the determinant

$$Z_L(t_1, t_2, t_3) = (\det{}' A_{k\ell})^{-\frac{1}{2}}, \tag{5.4}$$

where $t_i \equiv 2\ell_i/L$ with $t_1 + t_2 + t_3 = 1$, and the prime stands for removing the zero mode. This is explicitly evaluated in "discretized wave functional.nb" in two different ways that give the same answer.

In the limit $L \to \infty$, Z_L is divergent. There are two sources for the divergence. One is due to the discretization scheme and the normalization of the Gaussian measure, and should go like $\exp(const \cdot L)$. The other is due to the Weyl anomaly associated with the surface defined by the Strebel differential, and is universal. In particular, namely a cubic vertex of the ribbon graph, the metric goes like $ds^2 = e^{2\omega} dz d\bar{z} \sim |z| dz d\bar{z}$, and so $\omega \sim \frac{1}{2}\log|z|$. Need a puncture, on the otherhand, we have $\omega \sim -\log|z|$. The Weyl anomaly factor in going from the flat metric to the ribbon graph metric on the torus is e^{S_L} with the linear dilaton action S_L given simply by

$$S_L = \frac{1}{12\pi} \int_{T^2} d^2 z\, \partial\omega \bar{\partial}\omega. \tag{5.5}$$

As $\partial\omega \sim \frac{1}{4z}$ near a cubic vertex, and $\partial\omega \sim -\frac{1}{2z}$ near a puncture, we get logarithmic divergence

$$S_L \sim \frac{1}{12\pi} \cdot 2 \cdot 2\pi \left(2 \cdot \frac{1}{4^2} \log\frac{1}{\delta} + \frac{1}{2^2} \log\frac{1}{\epsilon}\right), \tag{5.6}$$

where δ and ϵ are the cutoff parameters near the pair of cubic vertex points and the puncture respectively. Note that the coefficient of the log divergence is independent from the moduli.

5.1 Torus one point function: $\partial X \bar{\partial} X$

Let's consider the torus one point function, for example of $\partial X \bar{\partial} X$. This is a weight $(1,1)$ primary operator, so

$$\langle : \partial X \bar{\partial} X : (0) \rangle_{T^2(t_1,t_2)} = |\widehat{f}(0)|^{-2} \langle : \partial X \bar{\partial} X : (0) \rangle_{D^2(t_1,t_2)}, \tag{5.7}$$

where the correlator on the RHS can be approximately computed as

$$\propto \int \left[\prod_{k=1}^{L} dX(k) \right] \Psi_{\partial X \bar{\partial} X}[X(k)] \delta_{\text{identification}}, \tag{5.8}$$

with $\delta_{\text{identification}}$ corresponding to the rules (5.2). The wavefunctional corresponding to the (normal ordered) operator $\partial X \bar{\partial} X$ inserted at the origin of the disk is given by Polchinski (2.8.28) and the paragraph below it as

$$\Psi_{\partial X \bar{\partial} X}[X(\sigma)] \propto (X_1 X_{-1} - \frac{\alpha'}{2}) \exp\left(-\frac{1}{\alpha'} \sum_{m \geq 1} m X_m X_{-m} \right). \tag{5.9}$$

The gaussian factor is the same as in (2.7), while the prefactor can be approximated in the discretization scheme as

$$X_1 X_{-1} \sim \frac{1}{L^2} \sum_{k,\ell=1}^{L} X(k) X(\ell) e^{\frac{2\pi i}{L}(k-\ell)}, \tag{5.10}$$

which we can denote after applying the identification rules as

$$X_1 X_{-1} \delta_{\text{identification}} \sim \sum_{k,\ell=1}^{L/2} C_{k\ell} X(k) X(\ell). \tag{5.11}$$

If we denote the shift used in the identification rule (5.2) as

$$s(n) = \begin{cases} \frac{L}{2} + \ell_1 + 1, & 1 \leq n \leq \ell_1 \\ \frac{L}{2} + \ell_2 + 2\ell_1 + 1, & \ell_1 + 1 \leq n \leq \ell_1 + \ell_2 \\ \frac{L}{2} + \ell_3 + 2\ell_2 + 2\ell_1 + 1, & \ell_1 + \ell_2 + 1 \leq n \leq \frac{L}{2} \end{cases} \tag{5.12}$$

then

$$C_{k\ell} = \frac{1}{L^2} \left(e^{\frac{2\pi i}{L}(k-\ell)} + e^{-\frac{2\pi i}{L}(k-\ell-s(k)+s(\ell))} + e^{\frac{2\pi i}{L}(k+\ell-s(\ell))} + e^{-\frac{2\pi i}{L}(k+\ell-s(k))} \right). \tag{5.13}$$

Finally, we wish to evaluate

$$\begin{aligned}\int \left[\prod_{k=1}^{L/2} dX(k)\right] \exp\left(-\sum_{k,\ell=1}^{L/2} A_{k\ell}X(k)X(\ell)\right) \sum_{r,s=1}^{L/2} C_{rs}X(r)X(s) \\ = (\det{}'A)^{-\frac{1}{2}}\frac{1}{2}\mathrm{tr}\,'\left(CA^{-1}\right).\end{aligned} \tag{5.14}$$

Recall that A has a zero eigenvector, namely $\sum_{k=1}^{L/2} X(k)$, so A^{-1} is undefined. However recall that $\sum_{k,\ell} X(k)X(\ell)A_{k\ell}$ is independent of this zero mode, and one can see that $\sum_{k=1}^{L/2} C_{k\ell} = 0 = \sum_{k=1}^{L/2} C_{\ell k}$ for any ℓ, so that $\sum_{k,\ell} X(k)X(\ell)C_{k\ell}$ is independent of this zero mode as well. Therefore we can consider the "reduced" $(\frac{L}{2}-1)\times(\frac{L}{2}-1)$ matrices $\widetilde{A}$ and $\widetilde{C}$ defined by imposing the additional constraint

$$X(\frac{L}{2}) = -\sum_{k=1}^{L/2} X(k) \tag{5.15}$$

after which $\widetilde{A}$ is invertible and

$$\begin{aligned}&\int \left[\prod_{k=1}^{L/2} dX(k)\right] \exp\left(-\sum_{k,\ell=1}^{L/2} A_{k\ell}X(k)X(\ell)\right) \sum_{r,s=1}^{L/2} C_{rs}X(r)X(s) \\ &= (\text{const.})\int \left[\prod_{k=1}^{L/2-1} dX(k)\right] \exp\left(-\sum_{k,\ell=1}^{L/2-1} \widetilde{A}_{k\ell}X(k)X(\ell)\right) \sum_{r,s=1}^{L/2-1} \widetilde{C}_{rs}X(r)X(s) \\ &= \sqrt{\frac{L}{2}}\,(\det\widetilde{A})^{-\frac{1}{2}}\frac{1}{2}\mathrm{tr}\,\left(\widetilde{C}\widetilde{A}^{-1}\right),\end{aligned} \tag{5.16}$$

with the proportionality factor $\frac{L}{2}\det{}'A = \det\widetilde{A}$ determined experimentally.

We can now consider

$$\begin{aligned}(Z(\tau(t_1,t_2)))^{-1}\langle :\partial X\bar{\partial}X:(0)\rangle_{T^2(t_1,t_2)} &= |\widehat{f}(0)|^{-2}\frac{\langle :\partial X\bar{\partial}X:(0)\rangle_{D^2(t_1,t_2)}}{\langle 1\rangle_{D^2(t_1,t_2)}} \\ &\sim \frac{1}{2}|\widehat{f}(0)|^{-2}\left(\mathrm{tr}\,(\widetilde{C}\widetilde{A}^{-1}) - \frac{\alpha'}{2}\right),\end{aligned} \tag{5.17}$$

where the determinants, and any Weyl anomaly factors should cancel between the numerator and denominator of the RHS calculated in the discretization scheme. Finally, from Polchinski (7.2.16) but with $\partial_{w'} \to \partial_{\bar{w}'}$, we find that

$$(Z(\tau))^{-1}\langle :\partial X\bar{\partial}X:(0)\rangle_{T^2(\tau)} = -4\pi^2\frac{\alpha'}{8\pi\tau_2}, \tag{5.18}$$

and would expect that

$$\mathrm{tr}\,(\widetilde{C}\widetilde{A}^{-1}) - \frac{1}{2\pi} = \frac{|\widehat{f}(0)|^2}{2\tau_2}, \tag{5.19}$$

which is modular invariant. In the above, the extra factor of $4\pi^2$ comes from the fact that Polchinski's torus has periods 2π, $2\pi\tau$ while our torus has periods 1, τ. We also make use of the fact that in our Mathematica implementation the convention is $\alpha' = \frac{1}{\pi}$.

We find that, to good numerical accuracy,

$$|\widehat{f}(0)|^{-2}\left(\frac{1}{2\pi} - \mathrm{tr}\,(\widetilde{C}\widetilde{A}^{-1})\right) = \frac{1}{2\tau_2} \tag{5.20}$$

5.2 Torus one point function: T

We can also consider the torus one point function of the stress tensor operator $T = -\frac{1}{\alpha'}\partial X \partial X$, which is not a primary operator, and obeys

$$\langle T(0)\rangle_{T^2(t_1,t_2)} = |\widehat{f}(0)|^{-2}\langle T(0) - \frac{c}{12}\{u,z\}|_{z=0}\rangle_{D^2(t_1,t_2)} \tag{5.21}$$

where u is the coordinate on the flat torus with periods 1 and τ, c is the central charge, and $\{u,z\}$ is the Schwarzian

$$\{u,z\}|_{z=0} = \frac{2\widehat{f}''\widehat{f} - 3(\widehat{f}')^2}{2\widehat{f}^2}\Big|_{z=0} = \frac{\widehat{f}''}{\widehat{f}}\Big|_{z=0} = f''(0). \tag{5.22}$$

In the above we make use of the fact that symmetry under rotation by π means only even powers of z appear in f and therefore all odd derivatives vanish at the origin, also that $\widehat{f}$ and f only differ by a constant rescaling, as well as that $f(0) = 1$.

From Polchinski (7.2.18), we read off that the LHS is given by

$$\langle T(0)\rangle_{T^2(t_1,t_2)} = -4\pi^2 Z(\tau(t_1,t_2))\left(\frac{1}{6}\frac{\partial_u^3\vartheta_1(0|\tau)}{\partial_u\vartheta_1(0|\tau)} + \frac{1}{8\pi\tau_2}\right) \tag{5.23}$$

where the additional factor of $4\pi^2$ is the same as explained under (5.19). Now the RHS we wish to calculate numerically. Let us introduce a generalization of the notation of 5.1,

$$\begin{aligned}(C_{m,n})_{k\ell} = \frac{|m|!|n|!}{L^2}\Big(&e^{-\frac{2\pi i}{L}(mk+n\ell)} + e^{\frac{2\pi i}{L}(mk+n\ell-ms(k)-ns(\ell))} \\ &+ e^{-\frac{2\pi i}{L}(mk-n\ell+ns(\ell))} + e^{\frac{2\pi i}{L}(mk-n\ell-ms(k))}\Big).\end{aligned} \tag{5.24}$$

where $C_{m,n}$ is the appropriate matrix to consider when the operator inserted at the origin is $:\partial^m X \partial^n X:$ (replace ∂^m with $\bar{\partial}^{|m|}$ if m is negative, and similarly for n) and $C = C_{1,-1}$.

Recalling that $\alpha' = 1/\pi$ in our convention used to define the matrix A in Mathematica, we find that to good numerical accuracy

$$4\pi^2 \left(\frac{1}{6} \frac{\partial_u^3 \vartheta_1(0|\tau)}{\partial_u \vartheta_1(0|\tau)} + \frac{1}{8\pi\tau_2} \right) = |\widehat{f}(0)|^{-2} \left(\pi \ \mathrm{tr}\, (\widetilde{C}_{1,1} \widetilde{A}^{-1}) + \frac{1}{12} f''(0) \right). \tag{5.25}$$

5.3 Compact Boson

Consider the torus partition function of a compact boson X with radius $X \sim X + 2\pi R$. On a unit circle, a configuration of X can be parametrized by the constant, the winding mode, and nonzero Fourier modes X_m:

$$X_b(\sigma) = x_0 + p\sigma R + \sum_{m=-\infty}^{\infty} X_m e^{im\sigma} \tag{5.26}$$

where $p \in \mathbb{Z}$ is the winding number. For any configuration with nonzero winding number, its classical action will be logarithmically divergent, and the wavefunctional $\Psi_1[X_b] = \int [\mathcal{D}X]_{X_b} e^{-S[X]}$ will be 0. Therefore, the wavefunctional $\Psi_1[X_b]$ is identical to that of a non-compact boson:

$$\Psi_1[p, X_m] \propto \delta_{p,0} \exp\left[-\frac{1}{\alpha'} \sum_{m \geq 1} m X_m X_{-m} \right] \tag{5.27}$$

which can be discretized as:

$$\begin{aligned} \Psi_1[X] &= \exp\left(-\frac{1}{\pi\alpha' L} \sum_{k,n=1}^{L} X(n) X(n+k) \frac{\sin \frac{\pi}{L}}{\cos \frac{2\pi k}{L} - \cos \frac{\pi}{L}} \right) \\ &\equiv \exp\left(-\frac{1}{\pi} \sum_{k,l=1}^{L} M_{kl} X(k) X(l) \right) \end{aligned} \tag{5.28}$$

The torus partition function can be computed as:

$$Z(l_1, l_2, l_3; R) = \sum_{s_1, s_2, s_3} \int [\mathcal{D}X] \Psi_1[X] \delta_{s_1, s_2, s_3}(...) \tag{5.29}$$

where the delta function imposes the following identification:

$$\begin{aligned} &X(L/2 + l_1 + 1 - n) = X(n) + 2\pi R s_1, \quad 1 \leq n \leq l_1, \\ &X(L/2 + 2l_1 + l_2 + 1 - n) = X(n) + 2\pi R s_2, \quad l_1 + 1 \leq n \leq l_1 + l_2, \\ &X(L + l_1 + l_2 + 1 - n) = X(n) + 2\pi R s_3, \quad l_1 + l_2 + 1 \leq n \leq L/2 \end{aligned} \tag{5.30}$$

This can be discretized as:

$$Z(l_i;R) = \int \prod_{i=1}^{L/2} dX(i) \exp\left[-\frac{1}{\pi}\left(\sum_{k,l=1}^{L/2} A_{kl}X(k)X(l) + 2\sum_{k=1}^{L/2} v_k X(k) + C\right)\right] \tag{5.31}$$

where:

$$\begin{aligned} A_{kl} &= M_{kl} + M_{k'l} + M_{kl'} + M_{k'l'} \\ v_k &= 2\pi R \sum_{i=1}^{3} s_i \sum_{m\in \text{seg } i} (M_{km'} + M_{k'm'}) \\ C &= 4\pi^2 R^2 \sum_{i,j}^{3} s_i s_j \sum_{k\in \text{seg } i} \sum_{m\in \text{seg } j} M_{k'm'} \end{aligned} \tag{5.32}$$

and the primed position k' stands for the position on the other half circle that's identified with k.

Now we explicitly perform the Gaussian integral. The matrix A has a zero mode given by $\sum_{k=1}^{L/2} X(k)$. Since

$$\begin{aligned} \sum_{k=1}^{L/2} v_k &= 2\pi R \sum_i s_i \sum_{k=1}^{L/2} \sum_{m\in \text{seg } i} (M_{km'} + M_{k'm'}) \\ &= 2\pi R \sum_i s_i \sum_{m\in \text{seg } i} \sum_{k=1}^{L} M_{km'} = 0 \end{aligned} \tag{5.33}$$

the shift v_k is then orthogonal to the zero mode, and hence the zero mode can be factored out, giving a factor of $2\pi R$. We factor out the zero mode by imposing the condition $X(L/2) = -\sum_{k=1}^{L/2-1} X(k)$, and the reduced matrix A' and the vector v' in the subspace of $X(1), ..., X(L/2)$ after this condition will be:

$$\begin{aligned} A'_{kl} &= A_{kl} - A_{k,L/2} - A_{L/2,m} + A_{L/2,L/2} \\ v'_k &= v_k - v_{L/2} \end{aligned} \tag{5.34}$$

and the Gaussian integral will become:

$$Z(l_i;R) = 2\pi R \int \prod_{i=1}^{L/2-1} dX(i) \exp\left[-\frac{1}{\pi}\left(\sum_{k,l=1}^{L/2-1} A'_{kl}X(k)X(l) + 2\sum_{k=1}^{L/2-1} v'_k X(k) + C\right)\right] \tag{5.35}$$

After performing the Gaussian integral in this subspace, the partition function is given

by:

$$
\begin{aligned}
Z(l_i;R) &= 2\pi R(\det A')^{-1/2} \sum_{s_1,s_2,s_3} \exp(-\pi^{-1}C + \pi^{-1} \sum_{k,l=1}^{L/2-1} v'_k (A')^{-1}_{kl} v'_l) \\
&= 2\pi R(\det A')^{-1/2} \sum_{s_1,s_2,s_3} \exp(-4\pi R^2 \sum_{i,j} s_i s_j T_{ij})
\end{aligned}
\tag{5.36}
$$

where the matrix T and W are defined as

$$
\begin{aligned}
T_{ij} &= \sum_{m\in \text{seg } i} \sum_{n\in \text{seg } j} \left(M_{m'n'} - \sum_{k,l=1}^{\frac{L}{2}-1} (A')^{-1}_{st} W_{km} W_{ln} \right) \\
W_{km} &= M_{km'} + M_{k'm'} - M_{\frac{L}{2},m'} - M_{(\frac{L}{2})',m'}
\end{aligned}
\tag{5.37}
$$

The three shifts s_1, s_2, s_3 store the same information with the 2 winding numbers around the 2 non-contractible cycles and the winding number around the puncture. Since the wavefunctional $\Psi_1[p, X_m]$ is nonzero only for configurations with winding number $p = 0$, we need to restrict onto the configurations with no winding around the puncture. For these configurations, the 3 shifts s_1, s_2, s_3 are not independent. We denote the cycle formed by going along segment 1 and then segment 2 as α, the one formed by going along segment 3 and then segment 1 backwards as β. The shifts s_i are then related to the winding numbers s_α, s_β around the α, β-cycles by

$$
s_1 = s_\alpha + s_\beta, \quad s_2 = s_\beta, \quad s_3 = -s_\alpha \tag{5.38}
$$

hence the three shifts are related by $s_3 = s_2 - s_1$. One can define the 2×2 matrix

$$
T'_{ij} = T_{ij} + (-1)^i T_{3j} + (-1)^j T_{i3} + (-1)^{i+j} T_{33} \tag{5.39}
$$

and the partition function can be written as

$$
\begin{aligned}
Z(l_i;R) &= 2\pi R(\det A')^{-1/2} \sum_{s_1,s_2} \exp(-4\pi R^2 \sum_{i,j=1}^{2} s_i s_j T'_{ij}) \\
&= 2\pi R(\det A')^{-1/2} \Theta(4iR^2 T')
\end{aligned}
\tag{5.40}
$$

where $\Theta(A)$ is the Siegel theta function of the matrix A. This can be evaluated using the builtin Mathematica function `SiegelTheta`[`A`, `0`] to evade explicit evaluation of infinite sums.

Since the $Z(l_i, R)$ only depend on R via the zero mode factor $2\pi R$ and the theta function, in order to check the T-duality relation $Z(l_i; R) = Z(l_i; R^{-1})$ (taking $\alpha' = 1$), we only need to check:

$$
R\Theta(4iR^2 T') = R^{-1}\Theta(4iR^{-2} T') \tag{5.41}
$$

and this has been checked numerically. We also checked numerically that the ratio between partition functions at different R and the same moduli matches with the analytical value:

$$\frac{Z(l_i;R_1)}{Z(l_i;R_2)} = \frac{Z_{\text{comp}}(\tau(l_i),R_1)}{Z_{\text{comp}}(\tau(l_i),R_2)}, \quad Z_{\text{comp}}(\tau,R) = \frac{1}{\sqrt{\tau_2}|\eta(\tau)|^2}\sum_{m,n} e^{-\frac{\pi R^2}{\tau_2}|m\tau-n|^2} \tag{5.42}$$

5.4 Compact boson: one point function on the disc in the discretized scheme

Let us now derive the discretized formula for the disk one point function of $:\partial X\partial X:(0)$. We continue to work in the reduced subspace obtained by imposing $X(\frac{L}{2}) = -\sum_{k=1}^{\frac{L}{2}-1} X(k)$, and denote the reduced variables by

$$X = (X(1),...,X(\tfrac{L}{2}-1))^T. \tag{5.43}$$

The physical winding lattice is two-dimensional, and in the notation used in the compact partition-function discussion above the zero puncture winding condition is

$$s_3 = s_2 - s_1. \tag{5.44}$$

Therefore it is more faithful to impose this relation from the start, and regard s_1, s_2 as the independent winding numbers. In a fixed sector (s_1, s_2), the compact partition function is then generated by the same reduced Gaussian used in the partition-function computation above,

$$\mathcal{Z}_{s_1,s_2} \propto \int d^{\frac{L}{2}-1}X \, \exp\left[-\frac{1}{\pi}\left(X^T A' X + 2(v')^T X + C\right)\right], \tag{5.45}$$

with $A'_{k\ell}$, v'_k and C defined in the compact partition-function discussion above. It is useful to rewrite the linear term as

$$v'_k = 2\pi R\sum_{i=1}^{2} U'_{ki}s_i, \qquad U'_{k1} = U_{k1} - U_{k3}, \qquad U'_{k2} = U_{k2} + U_{k3}, \qquad U_{ki} \equiv \sum_{m\in \text{seg}\, i} W_{km}, \tag{5.46}$$

and similarly

$$D_{ij} \equiv \sum_{m\in\text{seg}\, i}\sum_{n\in\text{seg}\, j} M_{m'n'}, \tag{5.47}$$

$$C = 4\pi^2 R^2 \sum_{i,j=1}^{2} s_i s_j D'_{ij}, \tag{5.48}$$

where D'_{ij} is defined from D_{ij} in exactly the same way as T'_{ij} was defined from T_{ij} in the partition-function discussion.

For the operator insertion, we start from the same X_1X_1 factor that appeared in the non-compact discussion of the stress tensor. After imposing the compact identification with shifts $2\pi Rs_i$ on the primed sites, the insertion takes the form

$$X_1 = \frac{1}{L}\sum_{m=1}^{L} X_{\text{full}}(m)e^{-\frac{2\pi i}{L}m}, \qquad X_{\text{full}} = PX + 2\pi RQs, \tag{5.49}$$

where P and Q are the same linear maps that reconstruct the full boundary data from the reduced variables X and the three segment shifts $s = (s_1, s_2, s_3)^T$. Therefore

$$X_1X_1 = \sum_{m,n=1}^{L} (N_{1,1})_{mn} X_{\text{full}}(m)X_{\text{full}}(n), \qquad (N_{1,1})_{mn} = \frac{1}{L^2}e^{-\frac{2\pi i}{L}(m+n)}. \tag{5.50}$$

Expanding $X_{\text{full}} = PX + 2\pi RQs$ gives

$$X_1X_1 = X^T(P^TN_{1,1}P)X + 4\pi R\, X^T(P^TN_{1,1}Q)s + 4\pi^2R^2\, s^T(Q^TN_{1,1}Q)s. \tag{5.51}$$

Thus the overall coefficient is fixed entirely by the discrete Fourier transform and the sewing identification, and there is no extra normalization beyond the factor L^{-2} already contained in $N_{1,1}$. After imposing $s_3 = s_2 - s_1$ and reducing to the two independent winding numbers, this becomes

$$X_1X_1\delta_{s_1,s_2} \sim X^T\widetilde{C}_{1,1}X + 4\pi R\sum_{i=1}^{2} X^TY'_{1,1;i}\, s_i + 4\pi^2R^2\sum_{i,j=1}^{2} s_is_jE'_{1,1;ij}. \tag{5.52}$$

Here $\widetilde{C}_{1,1}$ is the same reduced matrix that appeared in the non-compact discussion of the stress tensor, while $Y'_{1,1;i}$ and $E'_{1,1;ij}$ are obtained by substituting $s_3 = s_2 - s_1$ into the linear and quadratic shift-dependent terms respectively.

Therefore the numerator in the fixed winding sector (s_1, s_2) is, in the same normalization used above for the compact partition function,

$$\begin{aligned}\mathcal{N}_{s_1,s_2} = 2\pi R\int d^{\frac{L}{2}-1}X\left(X^T\widetilde{C}_{1,1}X + 4\pi R\sum_{i=1}^{2} X^TY'_{1,1;i}\, s_i + 4\pi^2R^2\sum_{i,j=1}^{2} s_is_jE'_{1,1;ij}\right)\\ \times\exp\left[-\frac{1}{\pi}\left(X^TA'X + 2(v')^TX + C\right)\right].\end{aligned} \tag{5.53}$$

Completing the square with $X = \widehat{X} - (A')^{-1}v'$, we obtain

$$\mathcal{N}_{s_1,s_2} = 2\pi R(\det A')^{-1/2}e^{-4\pi R^2\sum_{i,j=1}^{2} s_is_jT'_{ij}}\left[\frac{\pi}{2}\text{tr}\Big(\widetilde{C}_{1,1}(A')^{-1}\Big) - 4\pi R\sum_{i=1}^{2} s_i\, Y'^T_{1,1;i}(A')^{-1}v' + v'^T(A')^{-1}\widetilde{C}_{1,1}(A'\right. \tag{5.54}$$

where we used the expression for $\mathcal{Z}_{s_1,s_2}$ already obtained in the partition-function discussion. Dividing by $\mathcal{Z}_{s_1,s_2}$ then gives the normalized expectation value in that fixed sector,

$$\begin{aligned}\frac{\mathcal{N}_{s_1,s_2}}{\mathcal{Z}_{s_1,s_2}} =& \frac{\pi}{2}\mathrm{tr}\Big(\widetilde{C}_{1,1}(A')^{-1}\Big) - 4\pi R\sum_{i=1}^{2} s_i\, Y'^{T}_{1,1;i}(A')^{-1}v' \\ &+ v'^{T}(A')^{-1}\widetilde{C}_{1,1}(A')^{-1}v' + 4\pi^2R^2\sum_{i,j=1}^{2} s_i s_j E'_{1,1;ij} \\ =& \frac{\pi}{2}\mathrm{tr}\Big(\widetilde{C}_{1,1}(A')^{-1}\Big) + 4\pi^2R^2\sum_{i,j=1}^{2} s_i s_j \Xi'_{1,1;ij},\end{aligned} \tag{5.55}$$

where

$$\Xi'_{1,1;ij} = E'_{1,1;ij} - 2\sum_{k,\ell=1}^{\frac{L}{2}-1} Y'_{1,1;ki}(A')^{-1}_{k\ell}U'_{\ell j} + \sum_{k,\ell,m,n=1}^{\frac{L}{2}-1} U'_{ki}(A')^{-1}_{k\ell}\widetilde{C}_{1,1;\ell m}(A')^{-1}_{mn}U'_{nj}. \tag{5.56}$$

Again only the symmetric part of $\Xi'_{1,1;ij}$ contributes. This is the same matrix that would be obtained by first writing the answer in terms of the unconstrained three shifts and then substituting $s_3 = s_2 - s_1$. The physical normalized one point function is the ratio of the sector-summed numerator and denominator,

$$\frac{\langle :\partial X\partial X: (0)\rangle_{D^2,R}}{\langle 1\rangle_{D^2,R}} = \frac{\sum_{s_1,s_2}\mathcal{N}_{s_1,s_2}}{\sum_{s_1,s_2}\mathcal{Z}_{s_1,s_2}} = \frac{\sum_{s_1,s_2}\mathcal{Z}_{s_1,s_2}\left(\mathcal{N}_{s_1,s_2}/\mathcal{Z}_{s_1,s_2}\right)}{\sum_{s_1,s_2}\mathcal{Z}_{s_1,s_2}}, \tag{5.57}$$

not the sum of the individual ratios $\mathcal{N}_{s_1,s_2}/\mathcal{Z}_{s_1,s_2}$. Using the expression for $\mathcal{Z}_{s_1,s_2}$ already obtained in the partition-function discussion, we therefore arrive at

$$\frac{\langle :\partial X\partial X: (0)\rangle_{D^2,R}}{\langle 1\rangle_{D^2,R}} = \frac{\pi}{2}\mathrm{tr}\Big(\widetilde{C}_{1,1}(A')^{-1}\Big) + 4\pi^2R^2\frac{\sum_{s_1,s_2\in\mathbb{Z}}\left(\sum_{i,j=1}^{2} s_i s_j \Xi'_{1,1;ij}\right)e^{-4\pi R^2\sum_{i,j=1}^{2} s_i s_j T'_{ij}}}{\sum_{s_1,s_2\in\mathbb{Z}} e^{-4\pi R^2\sum_{i,j=1}^{2} s_i s_j T'_{ij}}}. \tag{5.58}$$

The first term is precisely the non-compact disk answer, while the second term is the compact correction coming from the nonzero mean of the shifted Gaussian in each winding sector. In the decompactification limit only the $s_1 = s_2 = 0$ sector survives, and (5.58) reduces to the non-compact result. Multiplying by $-\frac{1}{\alpha'}$ gives the corresponding disk one point function of the stress tensor.

5.5 Torus two point function of compact boson in flat coordinate

To compare with the discretized computation, it is useful to write the exact answer directly in the flat torus coordinate $w \sim w + 2\pi \sim w + 2\pi\tau$. We decompose the compact boson into

a single-valued part and a classical winding solution,

$$X(w,\bar{w}) = X_q(w,\bar{w}) + X_{\rm cl}^{(m,n)}(w,\bar{w}), \tag{5.59}$$

where in winding sector $(m,n) \in \mathbb{Z}^2$,

$$X_{\rm cl}^{(m,n)}(w+2\pi,\bar{w}+2\pi) = X_{\rm cl}^{(m,n)}(w,\bar{w}) + 2\pi R m, \tag{5.60}$$

$$X_{\rm cl}^{(m,n)}(w+2\pi\tau,\bar{w}+2\pi\bar{\tau}) = X_{\rm cl}^{(m,n)}(w,\bar{w}) + 2\pi R n. \tag{5.61}$$

The harmonic representative is

$$X_{\rm cl}^{(m,n)}(w,\bar{w}) = x_0 + \frac{R}{2i\tau_2}\left[(n-m\bar{\tau})w - (n-m\tau)\bar{w}\right], \tag{5.62}$$

so that

$$\partial X_{\rm cl}^{(m,n)} = \frac{R}{2i\tau_2}(n-m\bar{\tau}), \qquad \bar{\partial} X_{\rm cl}^{(m,n)} = -\frac{R}{2i\tau_2}(n-m\tau), \tag{5.63}$$

and the classical action is

$$S_{m,n} = \frac{\pi R^2}{\alpha'\tau_2}|n-m\tau|^2. \tag{5.64}$$

Accordingly, the compact boson partition function on the flat torus is, up to a moduli independent normalization factor,

$$Z_R(\tau,\bar{\tau}) \propto \frac{1}{\sqrt{\tau_2}|\eta(\tau)|^2}\Theta_R(\tau,\bar{\tau}), \tag{5.65}$$

with the genus one winding lattice sum

$$\Theta_R(\tau,\bar{\tau}) = \sum_{m,n\in\mathbb{Z}} e^{-S_{m,n}} = \sum_{m,n\in\mathbb{Z}} \exp\left[-\frac{\pi R^2}{\alpha'\tau_2}|n-m\tau|^2\right]. \tag{5.66}$$

Now consider the normalized torus two point function of ∂X,

$$\langle \partial X(w_1)\partial X(w_2)\rangle_{T^2,R} = \frac{\sum_{m,n}\int[\mathcal{D}X_q]\left(\partial X_q(w_1)+\partial X_{\rm cl}^{(m,n)}\right)\left(\partial X_q(w_2)+\partial X_{\rm cl}^{(m,n)}\right)e^{-S[X_q]-S_{m,n}}}{\sum_{m,n}\int[\mathcal{D}X_q]e^{-S[X_q]-S_{m,n}}}. \tag{5.67}$$

The cross term vanishes because the one point function of ∂X_q in the non-compact boson theory is 0. Hence

$$\langle \partial X(w_1)\partial X(w_2)\rangle_{T^2,R} = \langle \partial X(w_1)\partial X(w_2)\rangle_{T^2,\rm noncompact} + \frac{1}{\Theta_R(\tau,\bar{\tau})}\sum_{m,n\in\mathbb{Z}}\left(\partial X_{\rm cl}^{(m,n)}\right)^2 e^{-S_{m,n}}. \tag{5.68}$$

If we denote

$$\nu \equiv \frac{w_1 - w_2}{2\pi}, \tag{5.69}$$

then the first term is the usual zero mode subtracted propagator of the non-compact boson,

$$\langle \partial X(w_1)\partial X(w_2)\rangle_{T^2,\text{noncompact}} = \frac{\alpha'}{8\pi^2}\partial_\nu^2 \log \vartheta_1(\nu|\tau) + \frac{\alpha'}{8\pi\tau_2}, \tag{5.70}$$

which behaves as $-\frac{\alpha'}{2}(w_1 - w_2)^{-2} + O(1)$ as $w_1 \to w_2$.

For the second term, we use

$$\left(\partial X_{\text{cl}}^{(m,n)}\right)^2 = -\frac{R^2}{4\tau_2^2}(n - m\bar{\tau})^2. \tag{5.71}$$

Moreover,

$$\partial_\tau S_{m,n} = \frac{\pi i R^2}{2\alpha'\tau_2^2}(n - m\bar{\tau})^2, \tag{5.72}$$

so that

$$\partial_\tau \Theta_R(\tau,\bar{\tau}) = -\frac{\pi i R^2}{2\alpha'\tau_2^2}\sum_{m,n\in\mathbb{Z}}(n - m\bar{\tau})^2 e^{-S_{m,n}}. \tag{5.73}$$

It follows that

$$\frac{1}{\Theta_R(\tau,\bar{\tau})}\sum_{m,n\in\mathbb{Z}}\left(\partial X_{\text{cl}}^{(m,n)}\right)^2 e^{-S_{m,n}} = -\frac{i\alpha'}{2\pi}\partial_\tau \log \Theta_R(\tau,\bar{\tau}). \tag{5.74}$$

Therefore the exact compact boson two point function is

$$\langle \partial X(w_1)\partial X(w_2)\rangle_{T^2,R} = \frac{\alpha'}{8\pi^2}\partial_\nu^2 \log \vartheta_1(\nu|\tau) + \frac{\alpha'}{8\pi\tau_2} - \frac{i\alpha'}{2\pi}\partial_\tau \log \Theta_R(\tau,\bar{\tau}). \tag{5.75}$$

The dependence on $w_1 - w_2$ is identical to that of the non-compact boson; compactification only changes the constant term through the winding lattice sum. Taking the coincident point limit after subtracting the short distance singularity gives

$$\langle :\partial X\partial X: (w)\rangle_{T^2,R} = \frac{\alpha'}{24\pi^2}\frac{\vartheta_1'''(0|\tau)}{\vartheta_1'(0|\tau)} + \frac{\alpha'}{8\pi\tau_2} - \frac{i\alpha'}{2\pi}\partial_\tau \log \Theta_R(\tau,\bar{\tau}). \tag{5.76}$$

Compared with the non-compact boson, the compact boson has an additional moduli dependent constant. By comparing the OPE and the series expansion of the two point function at $z \to w$, we can determine the one point function of operators $:\partial^n X\partial X:$. Compared with the non-compact boson, the extra factor only contributes to the one point function of $:\partial X\partial X:$.

To compare with the discretized computation on the disc, let $u(z) = \int_0^z \widehat{f}(z')dz'$ be the normalized flat coordinate with periods $1, \tau$, and $w = 2\pi u$ the flat torus coordinate used above. Since ∂X has holomorphic weight 1 while $:\partial X \partial X:$ transforms with the Schwarzian anomaly, we have

$$\langle :\partial X \partial X: (0)\rangle_{D^2,R} = \left(\frac{dw}{dz}\Big|_{z=0}\right)^2 \langle :\partial X \partial X: (0)\rangle_{T^2,R} - \frac{\alpha'}{12}\{w,z\}\Big|_{z=0}. \tag{5.77}$$

Now

$$\frac{dw}{dz}\Big|_{z=0} = 2\pi \widehat{f}(0), \tag{5.78}$$

and because w and u differ only by a constant rescaling, $\{w,z\} = \{u,z\}$. By the π-rotation symmetry of the disc coordinate, only even powers of z appear in $f(z)$, so $f'(0) = 0$. Since $\widehat{f}$ differs from f only by an overall constant rescaling and $f(0) = 1$, we get

$$\{w,z\}\Big|_{z=0} = \frac{2\widehat{f}''\widehat{f} - 3(\widehat{f}')^2}{2\widehat{f}^2}\Big|_{z=0} = \frac{\widehat{f}''(0)}{\widehat{f}(0)} = f''(0). \tag{5.79}$$

Therefore

$$\langle :\partial X \partial X: (0)\rangle_{D^2,R} = 4\pi^2 \widehat{f}(0)^2 \langle :\partial X \partial X: (0)\rangle_{T^2,R} - \frac{\alpha'}{12} f''(0), \tag{5.80}$$

or equivalently

$$\langle :\partial X \partial X: (0)\rangle_{T^2,R} = \frac{\langle :\partial X \partial X: (0)\rangle_{D^2,R} + \frac{\alpha'}{12} f''(0)}{4\pi^2 \widehat{f}(0)^2}. \tag{5.81}$$

This is the relation that should be verified by the discretized computation on the disc.

6 Moduli and Weyl anomaly in disc coordinate

Let $z = e^{-iw}$ with the unit disc parameterized by $|z| < 1$ or $\mathrm{Im}(w) > 0$. The holomorphic 1-form on the torus can be expressed in the disc coordinate as a power series in z. By the expected divergence of $f(z \to z_i)$ for z_i the points on the boundary of the disk that map to the vertices of the ribbon graph in the torus, we can make a convenient ansatz

$$\begin{aligned} f(z)dz &= \frac{\sum_{n=1}^{\infty} a_n z^{n-1}}{(1-z^2)^{\frac{1}{3}}(1-z^2 e^{-4\pi i \frac{\ell_1}{L}})^{\frac{1}{3}}(1-z^2 e^{-4\pi i \frac{\ell_1+\ell_2}{L}})^{\frac{1}{3}}} dz \\ &= i \frac{\sum_{n=1}^{\infty} a_n e^{inw}}{(1-e^{2iw})^{\frac{1}{3}}(1-e^{2i(w-2\pi\frac{\ell_1}{L})})^{\frac{1}{3}}(1-e^{2i(w-2\pi i\frac{\ell_1+\ell_2}{L})})^{\frac{1}{3}}} dw. \end{aligned} \tag{6.1}$$

We can normalize with $a_1 = 1$, and solve $a_2, \cdots, a_L$ numerically by imposing the appropriate matching at discretized points

$$
\begin{aligned}
&\frac{\sum_{n=1}^{\frac{L}{2}} a_n e^{in(2\pi\frac{k-\frac{1}{2}}{L})}}{(1-e^{4i\pi\frac{k-\frac{1}{2}}{L}})^{\frac{1}{3}}(1-e^{4i\pi\frac{k-\frac{1}{2}-\ell_1}{L}})^{\frac{1}{3}}(1-e^{4i\pi\frac{k-\frac{1}{2}-\ell_1-\ell_2}{L}})^{\frac{1}{3}}} \\
&\quad = -\frac{\sum_{n=1}^{\frac{L}{2}} a_n e^{in(2\pi\frac{k'-\frac{1}{2}}{L})}}{(1-e^{4i\pi\frac{k'-\frac{1}{2}}{L}})^{\frac{1}{3}}(1-e^{4i\pi\frac{k'-\frac{1}{2}-\ell_1}{L}})^{\frac{1}{3}}(1-e^{4i\pi\frac{k'-\frac{1}{2}-\ell_1-\ell_2}{L}})^{\frac{1}{3}}}, \qquad 1 \le k \le \tfrac{L}{2}.
\end{aligned}
\tag{6.2}
$$

The relative sign is due to flipping the orientation on dw in the identification. The shift of k by $\frac{1}{2}$ is to put the Strebel vertices z_i, which are halfway between the discretized points at the edges of identified sections of the boundary, precisely at the angles in the ansatz. Evaluating the periods of $f(z)dz$ along α and β cycle then determines τ in terms of ℓ_1, ℓ_2, ℓ_3.

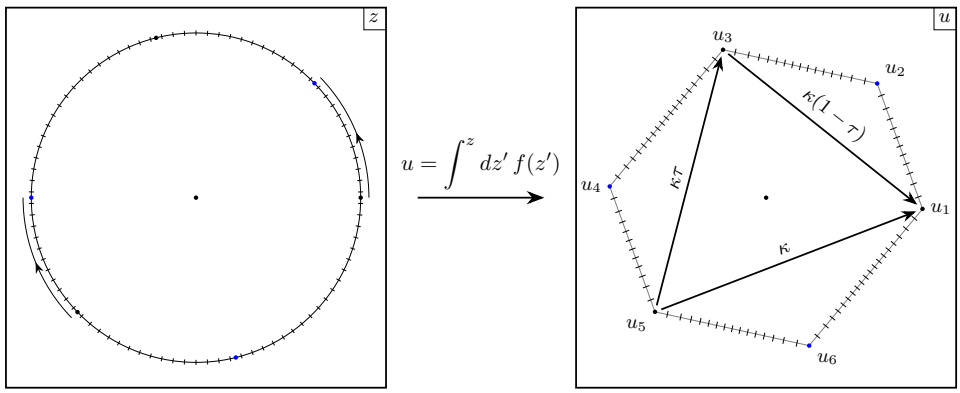

Figure 1: How $f(z)$ maps the circle to the torus, shown for the specific case of $\ell_i = \{11, 15, 19\}$. Note that the three black and three blue points correspond to one of the two Strebel vertices, respectively. The arcs in the z plane show how the sides of the cirle are identified with opposite orientation.

The flat coordinate on the torus (up to a constant rescaling) can be written as

$$
u = \int_{z_0}^{z} f(z')dz' \tag{6.3}
$$

The metric associated with the ribbon graph is

$$
\frac{dzd\bar{z}}{|z|^2} = \frac{dud\bar{u}}{|z\partial_z u|^2}, \tag{6.4}
$$

and so the Weyl factor is given by

$$
\omega = -\log|zf(z)|\,. \tag{6.5}
$$

(5.5) is evaluated as

$$\begin{aligned} S_L &= \frac{1}{12\pi}\int_D d^2z\partial_z\omega\bar{\partial}_{\bar{z}}\omega \\ &= \frac{1}{12\pi i}\oint_{\partial D} dz\,\omega\partial_z\omega, \end{aligned} \tag{6.6}$$

where D is the unit disc with the small neighborhoods of the singular points cut out. The divergent contribution from the component of ∂D near $z=0$ is moduli independent (as $f(0)=1$). Near one of the cubic vertex i.e. zero of Strebel differential, $u=u_i$ $(i=1,2)$, we have $\partial_z u = f(z) \approx b_i(u-u_i)^{-\frac{1}{2}}$, and $\partial_z f(z) = f(z)\partial_u f(z) \approx -\frac{1}{2}b_i^2(u-u_i)^{-2}$. It follows that

$$du\,\omega\partial_u\omega \approx -\frac{du}{4(u-u_i)}\log|f(z)| \tag{6.7}$$

near $u=u_i$. Therefore we have

$$S_L = \frac{\log|b_1|+\log|b_2|}{24} + \text{universal divergence.} \tag{6.8}$$

Here the cutoff is assumed to be $|u-u_i|>\delta$. But should we impose the cutoff on z instead? Near $u=u_i$, we have $z-z_i \approx \frac{2}{3b_i}(u-u_i)^{\frac{3}{2}}$, and $f(z) \approx b_i(\frac{3b_i}{2})^{-\frac{1}{3}}(z-z_i)^{-\frac{1}{3}} \equiv \nu_i(1-z/z_i)^{-\frac{1}{3}}$. If we impose the cutoff in z, which would seem to be the more compatible with the discretization scheme on the disc, the moduli dependent part of the Weyl anomaly would be given by

$$S_L = \frac{\log|b_1|+\log|b_2|}{36} + \text{universal divergence.} \tag{6.9}$$

6.1 Special case: $\ell_3 = 0$

In the special case when one of the $\ell_i = 0$ (we take $\ell_3 = 0$ without loss of generality), the corresponding torus has pure imaginary τ, and

$$f(z)dz = \frac{1}{(1-z^2)^{\frac{1}{2}}(1-z^2e^{-4i\pi\frac{\ell_1}{L}})^{\frac{1}{2}}}dz \tag{6.10}$$

exactly. Defining $\theta_1 \equiv 2\pi\frac{\ell_1}{L}$, the periods of f are given by

$$\begin{aligned} \alpha\text{-cycle: } & e^{i\frac{\theta_1}{2}}K(\cos\frac{\theta_1}{2}) \\ \beta\text{-cycle: } & ie^{i\frac{\theta_1}{2}}K(\sin\frac{\theta_1}{2}) \\ \tau =& i\frac{K(\sin\frac{\theta_1}{2})}{K(\cos\frac{\theta_1}{2})}. \end{aligned} \tag{6.11}$$

$K(k)$ is the complete elliptic integral of the first kind (`EllipticK[`k^2`]` in **Mathematica**). In addition $f(z) \approx \nu_i (z - z_i)^{-\frac{1}{2}}$, with

$$\nu_i = \frac{1}{\sqrt{\pm 4 i e^{\mp i\theta_1} \sin\theta_1}}. \tag{6.12}$$

Note that $\nu_1^2 + \nu_2^2 = \frac{1}{2}$ while $(|\nu_1| + |\nu_2|)^2 = \frac{1}{\sin(\theta_1)}$.

7 The bc system

The identity wave functional of bc system on the unit circle can be written as

$$\Psi_1[c(\sigma)] \propto \prod_{n=2}^{\infty} c_n, \tag{7.1}$$

where c_n is the n-th Fourier mode of $c(\sigma)$ (up to a factor of i due to the conformal transformation between cylinder and plane). In the disc coordinate, we can write $c(z) = \sum_{n\in\mathbb{Z}} c_n z^{1-n}$. It follows that $c(z)\Psi_1$ extends holomorphically to the interior of the unit disc, as expected. The analogous ground state wave functionals are

$$\Psi_c = c_1 \Psi_1 \propto \prod_{n=1}^{\infty} c_n, \quad \Psi_{c\partial c} = c_1 c_0 \Psi_1 \propto \prod_{n=0}^{\infty} c_n. \tag{7.2}$$

[Q: is there a way to make BRST symmetry manifest for the discretized matter+ghost system?]

Now discretizing the unit circle into L sites, we have the approximation

$$\Psi_1[c] \propto \prod_{m=2}^{\frac{L}{2}} \sum_{k=1}^{L} c(k) e^{2\pi i \frac{(m-1)k}{L}}. \tag{7.3}$$

Next, we identify pairs of $c(k)$'s according to $k' = s(k) - k$ for $s(k)$ defined in (5.12). Note that $c(z)$ should transform as a vector field so under $z \to z' = \frac{q}{z}$ for q a phase, $\frac{c(z)}{z} = -\frac{c(z')}{z'}$, i.e.

$$c(k') = -e^{2\pi i \frac{k'-k}{L}} c(k). \tag{7.4}$$

More explicitly

$$\begin{aligned}
&c(\tfrac{L}{2} + \ell_1 + 1 - n) = -e^{\frac{2\pi i}{L}(\frac{L}{2} + \ell_1 + 1 - 2n)} c(n), \qquad 1 \le n \le \ell_1, \\
&c(\tfrac{L}{2} + 2\ell_1 + \ell_2 + 1 - n) = -e^{\frac{2\pi i}{L}(\frac{L}{2} + 2\ell_1 + \ell_2 + 1 - 2n)} c(n), \qquad \ell_1 + 1 \le n \le \ell_1 + \ell_2, \\
&c(L + \ell_1 + \ell_2 + 1 - n) = -e^{\frac{2\pi i}{L}(L + \ell_1 + \ell_2 + 1 - 2n)} c(n), \qquad \ell_1 + \ell_2 + 1 \le n \le \tfrac{L}{2},
\end{aligned} \tag{7.5}$$

leaving $\frac{L}{2}$ independent variables $c(k)$, $k = 1, \cdots, \frac{L}{2}$. We then integrate the constrained $\Psi_1[c]$ over these $\frac{L}{2}$ independent variables. To obtain a nonzero result requires removing the constant mode of c.

Let us consider the constrained $\Psi_c[c]$, which can be put in the form

$$\prod_{m=1}^{\frac{L}{2}} \sum_{k=1}^{\frac{L}{2}} B_{mk} c(k). \tag{7.6}$$

The $\frac{L}{2} \times \frac{L}{2}$ matrix B is given by

$$\begin{aligned} B_{mn} &= e^{-2\pi i \frac{n}{L}} (e^{2\pi i \frac{mn}{L}} - e^{2\pi i \frac{m}{L}(\frac{L}{2}+\ell_1+1-n)}), \qquad 1 \le n \le \ell_1, \\ B_{mn} &= e^{-2\pi i \frac{n}{L}} (e^{2\pi i \frac{mn}{L}} - e^{2\pi i \frac{m}{L}(\frac{L}{2}+2\ell_1+\ell_2+1-n)}), \qquad \ell_1 + 1 \le n \le \ell_1 + \ell_2, \\ B_{mn} &= e^{-2\pi i \frac{n}{L}} (e^{2\pi i \frac{mn}{L}} - e^{2\pi i \frac{m}{L}(L+\ell_1+\ell_2+1-n)}), \qquad \ell_1 + \ell_2 + 1 \le n \le \tfrac{L}{2}. \end{aligned} \tag{7.7}$$

The result of integrating (7.6) over $c(k)$, $k = 1, \cdots, \frac{L}{2}$, is $\det B$.

The question however is what is the relation between $\det B$ and the torus correlator with appropriate b, c insertions. While there is a c insertion at the center of the disc, there should be a b insertion at the boundary of the disc. Moreover, the b insertion should be localized at the singular points on the boundary of disc where the different segments join. We are after the moduli integration measure

$$dt_1 dt_2 \langle \mathcal{B}_{t_1} \mathcal{B}_{t_2} c\widetilde{c} \rangle_{T^2(t_1,t_2)} \tag{7.8}$$

Transforming from the flat coordinate to the disc coordinate via (6.3), we end up with

$$dt_1 dt_2 \langle \mathcal{B}_{t_1} \mathcal{B}_{t_2} c\widetilde{c}(0) \rangle_{D^2(t_1,t_2)} |\widehat{f}(0)|^2, \tag{7.9}$$

where $\widehat{f}(z)$ is proportional to $f(z)$ of (6.1), normalized such that the period of $\widehat{f}(z)dz$ along the α-cycle is 1.

We find that, combining the bc system together with 26 free bosons X^μ, so that Weyl anomalies cancel, that in the disk frame,

$$|\det B| (\det{}' A)^{-13} \propto \tau_2^{-13} |\eta(\tau)|^{-48} |\widehat{f}(0)|^2 \frac{1}{3} \sum_{i=1}^{3} \left| \lim_{z \to z_i} (1 - \frac{z}{z_i})^{2/3} (\widehat{f}(z))^2 \right|^2. \tag{7.10}$$

In other words, $|\det B|$ calculates the $(\sum_i b(z_i) \widetilde{b}(\bar{z}_i)) c\widetilde{c}(0)$ correlator on the disk with $c\widetilde{c}$ inserted at the origin, and b and $\widetilde{b}$ inserted at the Strebel vertices z_i, in the form $\lim_{z \to z_i} (1 - \frac{z}{z_i})^{2/3} b(z)$, such that when one considers the transformation to $b(u)$ in the torus coordinate,

$(1-\frac{z}{z_i})^{2/3}$ cancels the divergence of the factors of $f(z)$ that are picked up. Note that in the special case where one of the $\ell_i = 0$, the power must be adjusted to $(1-\frac{z}{z_i})$.

In the language of 6, and denoting the α-cycle period of $f(z)$ as κ, so $\widehat{f}(z) = \frac{1}{\kappa} f(z)$, we can write

$$|\det B|(\det{}' A)^{-13} \propto \tau_2^{-13} |\eta(\tau)|^{-48} \frac{1}{|\kappa|^2} |\nu|^4 , \tag{7.11}$$

for ν the average ν_i.

7.1 Dependence on c position

Instead of inserting the c ghost at $z = 0$ in the disk frame, which gives $\Psi_c[c]$, we can consider $c(z) = \sum_{n\leq 1} c_n z^{1-n}$ inserted at a generic $|z| < 1$ and the corresponding $\Psi_{c(z)}[c]$, which will be built from

$$\Psi_{c_n}[c] \propto \left(\sum_{k=1}^{L} c(k) e^{2\pi i \frac{nk}{L}} \right) \prod_{m=2}^{\frac{L}{2}} \left(\sum_{k=1}^{L} c(k) e^{2\pi i \frac{mk}{L}} \right) , \tag{7.12}$$

with the same identification rules (7.5) leaving the usual $\frac{L}{2}$ independent $c(k)$ variables, and resulting in an $\frac{L}{2} \times \frac{L}{2}$ matrix determinant once they are integrated over. We denote this determinant $\det B^{(n)}$, with $\det B^{(1)} = \det B$ of the preceding section.

Clearly, $\det B^{(0)} = 0$ because under (7.5), the matrix has a zero row. Numerically, we also find that $\det B^{(-n)} = 0$ for even n **[because of the symmetry of the disk under rotation by π]**. In addition, we find that in the case where all three side lengths ℓ_i are equal, $\det B^{(n)} = 0$ for all $n \neq 1 - 6k$ for some $k \in \mathbb{Z}_{\geq 0}$, corresponding to the extra $\mathbb{Z}_6$ rotational symmetry of the disk frame in this special case.

We can define

$$g(z) = 1 + \sum_{n=1}^{L/2}{}' \det B^{(-n)} z^{1+n} \tag{7.13}$$

with the $\sum'$ indicating summation over odd n. $g(z)$ is proportional to the path integral of $\Psi_{c(z)}[c]$. Considering that the boundary of the unit disc is identified under $z \to z' = q/z$, with $|q| = 1$ being three different phases for the three different segments on the boundary, it turns out (numerically tested) that

$$\frac{g(z')}{dz'} = \frac{g(z)}{dz} \quad \Leftrightarrow \quad -\frac{g(z'(z))}{z'} = \frac{g(z)}{z}, \qquad \text{for } |z| = 1. \tag{7.14}$$

Therefore

$$g(z) = (f(z))^{-1}, \tag{7.15}$$

and $g(z)$ precisely corresponds to the z-dependent conformal transformation factor of $c\widetilde{c}(z,\bar{z})$ to the correlator on the disc. The fact that there is seemingly no other z-dependence on the position of $c\widetilde{c}$ implies that there is no extra ghost number 0 bc insertions along the boundary of the disc, besides possibly at the points corresponding to the vertices themselves, in the correlator calculated by the discretized path integral of $\Psi_{c(z)}[c]$.

7.2 Matter and bc-ghost on a genus-2 surface

One can parameterize the genus-2 amplitude via the period matrix Ω as

$$\frac{1}{(\det \mathrm{Im}\Omega)^{13}} \frac{1}{|\chi_{10}(\Omega)|^2}, \tag{7.16}$$

where $\chi_{10}(\Omega)$ is the Igusa-cusp form and Ω is the period matrix. In the disc frame, the matter +b-c determinant is computed, analogously to the genus 1 case, as

$$|\det B|(\det A')^2 \tag{7.17}$$

Generalizing the genus-1 formula to the case of genus 2, where there are two independent holomorphic one-forms on the boundary, a reasonable guess for the analytic formula of the numerical amplitude in equation 7.17 is

$$\frac{1}{(\det \mathrm{Im}\Omega)^{13}} \frac{1}{|\chi_{10}(\Omega)|^2} \tag{7.18}$$

8 Free fermion

The ground-state wavefunction for the free fermion can be written as

$$\Psi_1[\psi_r] \propto \prod_{r=\frac{1}{2}}^{\frac{L-1}{2}} \sum_{k=1}^{L} \psi(k) e^{2\pi i \frac{rk}{L}} \tag{8.1}$$

The fermion is a weight-$\frac{1}{2}$ primary, so under sewing together opposite sides of the circle, with $\theta \mapsto -\theta$, the ψ are identified as

$$\begin{aligned} &\psi(\tfrac{L}{2}+\ell_1+1-n) = \pm i\psi(n), \quad 1 \le n \le \ell_1, \\ &\psi(\tfrac{L}{2}+2\ell_1+\ell_2+1-n) = \pm i\psi(n), \quad \ell_1+1 \le n \le \ell_1+\ell_2, \\ &\psi(L+\ell_1+\ell_2+1-n) = \pm i\psi(n), \quad \ell_1+\ell_2+1 \le n \le \tfrac{L}{2}, \end{aligned} \tag{8.2}$$

where the choice of sign corresponds to the choice of NS/R boundary conditions along the α, β cycles.

The constrained $\Psi_1[\psi_r]$ can be put in the form

$$\prod_{m=1}^{\frac{L}{2}} \sum_{k=1}^{\frac{L}{2}} D_{mk}\psi(k), \tag{8.3}$$

where the matrix D is given by

$$\begin{aligned}
D_{mn} &= \exp\left(\tfrac{2\pi i}{L}(m-\tfrac{1}{2})n\right) \pm i \exp\left(\tfrac{2\pi i}{L}(m-\tfrac{1}{2})(\tfrac{L}{2}+\ell_1+1-n)\right), \quad 1 \le n \le \ell_1 \\
D_{mn} &= \exp\left(\tfrac{2\pi i}{L}(m-\tfrac{1}{2})n\right) \pm i \exp\left(\tfrac{2\pi i}{L}(m-\tfrac{1}{2})(\tfrac{L}{2}+2\ell_1+\ell_2+1-n)\right), \quad \ell_1+1 \le n \le \ell_1+\ell_2 \\
D_{mn} &= \exp\left(\tfrac{2\pi i}{L}(m-\tfrac{1}{2})n\right) \pm i \exp\left(\tfrac{2\pi i}{L}(m-\tfrac{1}{2})(L+\ell_1+\ell_2+1-n)\right), \quad \ell_1+\ell_2+1 \le n \le \frac{L}{2}.
\end{aligned} \tag{8.4}$$

Analagously to the bc-ghost correlator, the result of integrating the fermion vacuum wavefunctional over ψ_r is $|\det D|$.

The NS-NS sector amplitude on the flat torus is

$$\left|\frac{\theta_3(0,\tau)}{\eta(\tau)}\right|^2 \tag{8.5}$$

with similar contributions from the NS/R and R/NS sectors. The Weyl anomaly is 1/2 the Weyl anomaly of the Empirically, one finds that, proceeding counterclockwise along the bottom segment of the disc, NS/NS corresponds to $-i, -i, +i$, NS/R corresponds to $-i, +i, -i$, and R/NS corresponds to $+i, -i, -i$.

The R/R sector should evaluate to zero without an insertion of the Ramond sector zero mode, which we have not yet inserted.

9 Higher Genus

9.1 Compact boson at higher genus

For genus 2 surface with 1 puncture, the ribbon graph has 1 face, 6 vertices and 9 edges. There are 4 different topologies of graphs that covers different region of $\mathcal{M}_{2,1}$. Given a ribbon graph, we now describe the algorithm of computing the compact boson partition function on the Riemann surface associated with that ribbon graph. Although we use genus 2 as example, this can be trivially generalized to the case of arbitrary genera.

In order to write down the identification map, Let's label the segments S_i $(i = 1, ..., 18)$ from $k = 1$ with increasing k. The edges can be written as E_a $(a = 1, ..., 9)$, where each E_a pairs 2 segments $S_{i_1(a)}, S_{i_2(a)}$ (we always take $i_1(a) < i_2(a)$) and glue them in the opposite orientation. We denote the edge associated with segment S_i by $E_{a(i)}$ and its length $\ell_{a(i)}$. The identification map of compact boson on edge E_a is then:

$$X\left(\sum_{j\leq i_1(a)} \ell_{a(j)} + \sum_{k<i_2(a)} \ell_{a(k)} - k + 1\right) = X(k) + 2\pi R s_a, \quad \sum_{j<i_1(a)} \ell_{a(j)} + 1 \leq k \leq \sum_{j\leq i_1(a)} \ell_{a(j)} \tag{9.1}$$

One additional complication is that, for higher genus, the independent variables are not simply $X(k)$ with $1 \leq k \leq L/2$, since the segments in the same half-circle can be glued together. We take the set of independent variables to be $\mathcal{K} = \cup_{a=1}^{9}\{k, k \in S_{i_1(a)}\}$, and denote $\mathcal{K}' = \mathcal{K}\backslash\{1\}$.

The condition of no winding around the unit circle can be translated to the following 6 constraints: For each vertex V_A, we label the segment that connects to it with the smallest label $S_{i_0(A)}$ (which includes $S_0 = S_{18}$), and define

$$i_1(A) = i_0(A) + 1, \quad i_2(A) = p(i_1(A)) + 1, \quad i_3(A) = p(i_2(A)) + 1 \tag{9.2}$$

where $S_{p(i)}$ is the edge paired with S_i. The constraints are given by:

$$\mathrm{sgn}(i_1(A))s_{a(i_1(A))} + \mathrm{sgn}(i_2(A))s_{a(i_2(A))} + \mathrm{sgn}(i_3(A))s_{a(i_3(A))} = 0 \tag{9.3}$$

where $\mathrm{sgn}(i)$ is defined such that $\mathrm{sgn}(i) = 1$ if $i < p(i)$ and -1 if $i > p(i)$. In a graph theory way, this can be repacked into a matrix equation

$$D_{A,a}s_a = 0 \tag{9.4}$$

where $s \in \mathbb{Z}^9$ is the vector of 9 shifts and $D \in \mathbb{Z}^{6\times 9}$. The independent shifts then lives in the kernel $\ker_{\mathbb{Z}}(D)$. If we orient each edge E_a in the counter-clockwise direction of $S_{i_1(a)}$, then the matrix D is then given by the oriented incidence matrix:

$$D_{A,a} = \begin{cases} 1 & \text{if edge } a \text{ into vertex A} \\ -1 & \text{if edge } a \text{ out of vertex A} \end{cases} \tag{9.5}$$

and the dimension of $\ker(D)$ is given by $E - V + 1 = 2g$. Namely, among the 6 constraints, only 5 of them are independent. This reduces the independent number of shifts from 9 to 4, which corresponds to the holonomies around the 4 non-contractible 1-cycles.

A basis of $\ker(D)$ can be built in the following way: One first choose a spanning tree of the oriented graph, which contains $V - 1 = 5$ edges. For each of the 4 edges $e = (A_1(e), A_2(e))$

that's not in the spanning tree, one can find a cycle C_e by starting from $A_1(e)$, going through e to get to $A_2(e)$, then take the unique path in the spanning tree from $A_2(e)$ back to $A_1(e)$. From this cycle C_e, one then define a vector

$$(b_e)_a = \begin{cases} 1 & \text{if edge } a \text{ is traversed with same orientation in } C_e \\ -1 & \text{if edge } a \text{ is traversed with opposite orientation in } C_e \\ 0 & \text{if edge } a \text{ not in } C_e \end{cases} \tag{9.6}$$

and the 4 b_e's will form a integral basis of $\ker(D)$. If we define the matrix $B \in \mathbb{Z}^{9\times 4}$ by $B_{a,e} = (b_e)_a$, then an arbitrary vector in $\ker(D)$ can be represented by Bn, where $n \in \mathbb{Z}^4$ is a lattice vector in the 4-dimensional integral lattice.

The genus-2 compact boson partition function can be discretized as:

$$Z^{(2)}(\ell_a; R) = 2\pi R \int \prod_{k\in\mathcal{K}} dX(k) \exp\left[-\frac{1}{\pi}\left(\sum_{k,l\in\mathcal{K}'} A'_{kl} X(k)X(l) + 2\sum_{k\in\mathcal{K}'} v'_k X(k) + C\right)\right] \tag{9.7}$$

where, similar to (5.32),

$$\begin{aligned} A_{kl} &= M_{kl} + M_{k'l} + M_{kl'} + M_{k'l'} \\ A'_{kl} &= A_{kl} - A_{1,l} - A_{k,1} - A_{1,1} \\ v_k &= 2\pi R \sum_{a=1}^{9} s_a \sum_{l\in S_{i_1(a)}} (M_{kl'} + M_{k'l'}) \\ v'_k &= v_k - v_1 \\ C &= 4\pi^2 R^2 \sum_{a,b=1}^{9} s_a s_b \sum_{k\in S_{i_1(a)}} \sum_{l\in S_{i_1(b)}} M_{k'm'} \end{aligned} \tag{9.8}$$

Following the same procedure with the genus-1 case, the partition function can be computed as:

$$Z^{(2)}(\ell_a; R) = 2\pi R (\det A')^{-1/2} \Theta(4iR^2 T') \tag{9.9}$$

where T' is the following matrix T_{ab} restricted to the 4-dimensional subspace.

$$\begin{aligned} T_{ab} &= \sum_{k\in S_{i_1}(a)} \sum_{l\in S_{i_1}(b)} \left(M_{k'l'} - \sum_{m,n\in\mathcal{K}'} (A')^{-1}_{mn} W_{mk} W_{nl} \right) \\ W_{km} &= M_{km'} + M_{k'm'} - M_{1,m'} - M_{1',m'} \end{aligned} \tag{9.10}$$

The restriction from T to T' can be performed by simply taking

$$T' = B^T T B \tag{9.11}$$

where the matrix B^T is the one that maps $\mathbb{Z}^9$ to the constrained subspace $\ker(D)$.

At genus g, the T-duality transformation of compact boson partition function is given by

$$Z^{(g)}(R) = R^{2-2g} Z^{(g)}(R^{-1}) \tag{9.12}$$

where the additional factor R^{2-2g} comes from the dilaton shift in the Buscher rules. In terms of the matrix T', this reads

$$R^g \Theta(4iR^2 T') = R^{-g} \Theta(4iR^{-2} T') \tag{9.13}$$

which has been tested numerically for all 4 topologies of genus 2 and all 818 topologies of genus 3. For genus 4, enumerating and storing all ribbon graph data becomes a problem, but T-duality has been checked for a single topology.

For genus g, the compact boson partition function $Z^{(g)}(\Omega, R)$ can be expressed in a lattice sum involving the period matrix Ω:

$$Z^{(g)}_{\text{comp}}(\Omega, R) = 2\pi R Z^{(g)}_{\text{bos}}(\Omega) \sum_{m,n\in\mathbb{Z}^g} \exp\left(-\pi R^2 (m+\Omega n)^T (\operatorname{Im}\Omega)^{-1} (m+\bar{\Omega} n)\right) \tag{9.14}$$

where the free boson partition function $Z_{\text{bos}}(\Omega)$ does not depend on the radius R. In term of T',

$$\frac{\Theta(4iR_1^2 T')}{\Theta(4iR_2^2 T')} = \frac{\sum_{m,n\in\mathbb{Z}^g} \exp\left(-\pi R_1^2 (m+\Omega n)^T (\operatorname{Im}\Omega)^{-1} (m+\bar{\Omega} n)\right)}{\sum_{m,n\in\mathbb{Z}^g} \exp\left(-\pi R_2^2 (m+\Omega n)^T (\operatorname{Im}\Omega)^{-1} (m+\bar{\Omega} n)\right)} \tag{9.15}$$

and this has been checked for genus 2 using codes in `compact_partition.py` and `ell_to_tau.py`.

L-dependent Renormalization of $Z(\ell_a; R)$

The lattice constructed provided us a *regularization* scheme, and if we want to have a finite answer when taking the $L \to \infty$ limit, we need to further *renormalize* the action by adding counterterms. For the theory at self-dual radius $R = 1$, we found numerically that the large-L behavior of $Z^{(g)}(\ell_a; R)$ is given by:

$$\log Z^{(g)}(L; \Omega; R = 1) = c(\Omega) + \gamma L + \alpha(g) \log L \tag{9.16}$$

where $c(\Omega)$ is the moduli-dependent finite part, $\gamma, \alpha(g)$ are both independent of the moduli. $\gamma = \frac{1}{4}\log 2$ is a constant while $\alpha(g)$ is given by the following:

$$\begin{aligned} \alpha(1) &= -1.13889 \\ \alpha(2) &= -1.41667 \\ \alpha(3) &= -1.69445 \\ \alpha(4) &= -1.97222 \\ \alpha(5) &= -2.24993 \end{aligned} \tag{9.17}$$

which suggests that $\alpha(g)$ has a linear dependence in g:

$$\alpha(g) = \alpha_0 + \alpha_1 g, \quad \alpha_0 = -0.86112, \quad \alpha_1 = -0.27778 \tag{9.18}$$

One can also compute the same large-L dependence for theories with $R \neq 1$ and linear fit the parameters $\alpha(g)$. For theories with different R, both the slope and the intercept of $\alpha(g)$ depends on R, but γ stays the same around $\frac{1}{4}\log 2$.

In order to arrive at a finite partition function in the $L \to \infty$ limit, we choose the renormalization scheme by adding the following counterterms to the (Euclidean) compact boson action:

$$S_{\text{counterterm}} = \gamma L + (\alpha_0 + \alpha_1)\log L - \frac{\alpha_1}{8\pi}\log L \int_\Sigma \mathrm{d}^2\sigma\sqrt{g}\, R(g) \tag{9.19}$$

and the partition function in this renormalization scheme is given by simply $\exp c(\Omega)$.

The analytical value $Z^{(g)}_{\text{comp}}$ of the compact boson partition function is computed in a different scheme and also a different Weyl frame. One can check numerically that, at genus 1, after taking the Weyl anomaly factor into account, the renormalized partition function $\exp c(\tau)$ and the analytic value $Z^{(1)}_{\text{comp}}(\tau)$ only differs by a finite renormalization of the cosmological constant, namely:

$$\frac{\exp(c(\tau) - S_L(\tau))}{Z^{(1)}_{\text{comp}}(\tau)} = \text{ Const.} \tag{9.20}$$

where $S_L(\tau)$ is given by (6.9).

10 Analytic construction

10.1 Trinion vertex

The trinion vertex with the three faces of equal circumference 2π is constructed as

$$\{\Psi_1 \otimes \Psi_2 \otimes \Psi_3\}_{[1,1,1]} = \left\langle \prod_{i=1}^{3} [\Psi_i(0)]^{f_i} \right\rangle, \tag{10.1}$$

where f_i are the conformal maps

$$f_i = h^{-1} \circ R_i \circ h, \tag{10.2}$$

with

$$h(z) \equiv \frac{1+iz}{1-iz} \tag{10.3}$$

and

$$R_1(w) = e^{\frac{2\pi i}{3}} w^{\frac{2}{3}}, \qquad R_2(w) = w^{\frac{2}{3}}, \qquad R_1(w) = e^{-\frac{2\pi i}{3}} w^{\frac{2}{3}}. \tag{10.4}$$

This construction is analogous to Witten's OSFT cubic vertex. Note however that Ψ_i are not necessarily closed string fields, in that they are not required to obey level matching condition or being annihilated by b_0^-.

More generally, we need to consider trinion vertex of face circumferences $2\pi p_i$, $i = 1, 2, 3$, $p_i > 0$. Up to permutation, there are two qualitatively distinct cases: (1) p_1, p_2, p_3 obey triangular inequality, in which case the ribbon graph is the theta graph, where the three edge lengths ℓ_1, ℓ_2, ℓ_3 are related by

$$2\pi p_1 = \ell_2 + \ell_3, \qquad 2\pi p_2 = \ell_3 + \ell_1, \qquad 2\pi p_3 = \ell_1 + \ell_2, \tag{10.5}$$

and (2) p_1, p_2, p_3 do not obey triangular inequality, in which case the ribbon is the dumbbell graph, where the three edge lengths are related by

$$2\pi p_1 = \ell_1, \qquad 2\pi p_2 = \ell_2, \qquad 2\pi p_3 = \ell_1 + \ell_2 + 2\ell_3. \tag{10.6}$$

We will denote the general trinion vertex by

$$\{\Psi_1 \otimes \Psi_2 \otimes \Psi_3\}_{[p_1,p_2,p_3]}. \tag{10.7}$$

To construct the explicit conformal map, we use the Strebel differential

$$\omega = \frac{P(z)}{(z-z_1)^2(z-z_2)^2(z-z_3)^2} dz^2, \tag{10.8}$$

where $P(z)$ is a quadratic polynomial in z determined by the conditions

$$\frac{P(z_1)}{z_{12}^2 z_{13}^2} = -p_1^2, \qquad \frac{P(z_2)}{z_{21}^2 z_{23}^2} = -p_2^2, \qquad \frac{P(z_3)}{z_{31}^2 z_{32}^2} = -p_3^2. \tag{10.9}$$

The relevant conformal maps f_i are given by

$$f_i^{-1}(z) = \exp\left(\frac{1}{p_i}\int^z \sqrt{\omega}\right), \qquad z \in D_i, \tag{10.10}$$

where D_i are domains of disc topology that contain z_i, such that f_i maps the unit circle to ∂D_i. We can then write

$$\{\Psi_1 \otimes \Psi_2 \otimes \Psi_3\}_{[p_1,p_2,p_3]} = \left\langle \prod_{i=1}^{3} [\Psi_i(z_i)]^{f_i} \right\rangle. \tag{10.11}$$

10.2 General correlator associated with a ribbon graph

A general ribbon graph of genus g and n faces can be decomposed into $2g-2+n$ trinion graphs, stitched together with twists. The corresponding correlators can be determined by tracing over internal states with twists.

For instance, the $(g,n)=(0,4)$ case is given by

$$\{\Psi_1\otimes\Psi_2\otimes\Psi_3\otimes\Psi_4\}^{(0,4)}_{[p_1,p_2,p_3,p_4;x,y]}=\sum_I\{\Psi_1\otimes\Psi_2\otimes e^{2\pi i L_0^- y}\phi_I\}_{[p_1,p_2,x]}\{\phi_I^\vee\otimes\Psi_3\otimes\Psi_4\}_{[x,p_3,p_4]}, \tag{10.12}$$

where ϕ_I and $\phi_I^\vee$ are dual bases (with respect to BPZ inner product) of the worldsheet CFT state space. The underlying ribbon graph has 4 vertices, 6 edges, 4 faces.

For (p_1,p_2,x) and (p_3,p_4,x) obeying triangular inequalities, the 6 edge lengths ℓ_i are related by

$$\begin{aligned}
2\pi p_1&=\ell_1+\ell_2+\ell_3,\\
2\pi p_2&=\ell_1+\ell_4+\ell_5,\\
2\pi p_3&=\ell_3+\ell_4+\ell_6,\\
2\pi p_4&=\ell_2+\ell_5+\ell_6,\\
2\pi x&=\ell_2+\ell_3+\ell_4+\ell_5,\\
2\pi y&=\ell_2.
\end{aligned} \tag{10.13}$$

In other cases, the underlying ribbon graph may also involve a dumbbell subgraph.

As another example, for $(g,n)=(1,1)$, we have

$$\{\Psi\}^{(1,1)}_{[p;x,y]}=\sum_I\{\Psi\otimes e^{2\pi i L_0^- y}\phi_I\otimes\phi_I^\vee\}_{[p,x,x]}. \tag{10.14}$$

Here $p>2x$. The underlying ribbon graph has 2 vertices, 3 edges, 1 face. It is formed out of self-plumbing of a dumbell graph, whose edge lengths $\widetilde{\ell}_1,\widetilde{\ell}_2,\widetilde{\ell}_3$ are related by

$$2\pi p=2\widetilde{\ell}_1+\widetilde{\ell}_2+\widetilde{\ell}_3,\qquad 2\pi x=\widetilde{\ell}_2=\widetilde{\ell}_3. \tag{10.15}$$

The $(g,n)=(1,1)$ ribbon graph has three edges of length ℓ_1,ℓ_2,ℓ_3, related by

$$\ell_1=\widetilde{\ell}_1,\qquad \ell_2=y,\qquad \ell_3=\widetilde{\ell}_2-y. \tag{10.16}$$

Note that $\ell_1+\ell_2+\ell_3=\widetilde{\ell}_1+\widetilde{\ell}_2=\pi p$.